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Case #1

A 65-year-old **black** male with **history of type 2 diabetes mellitus** and **osteoarthritis** presents to the clinic with a **complaint of headaches**. He currently takes **ibuprofen and metformin**. The patient has a **family history of heart failure and hypertension (HTN)**. He lives a **sedentary** lifestyle and **rarely exercises**. His **job is stressful, and he works long hours,** so his diet consists of mostly **fast food**. He has been **smoking** for 35 years and has two **beers** a night. He says he measures his blood pressure at home, and it is **140/80 mmHg**. He brought his cuff in with him today.

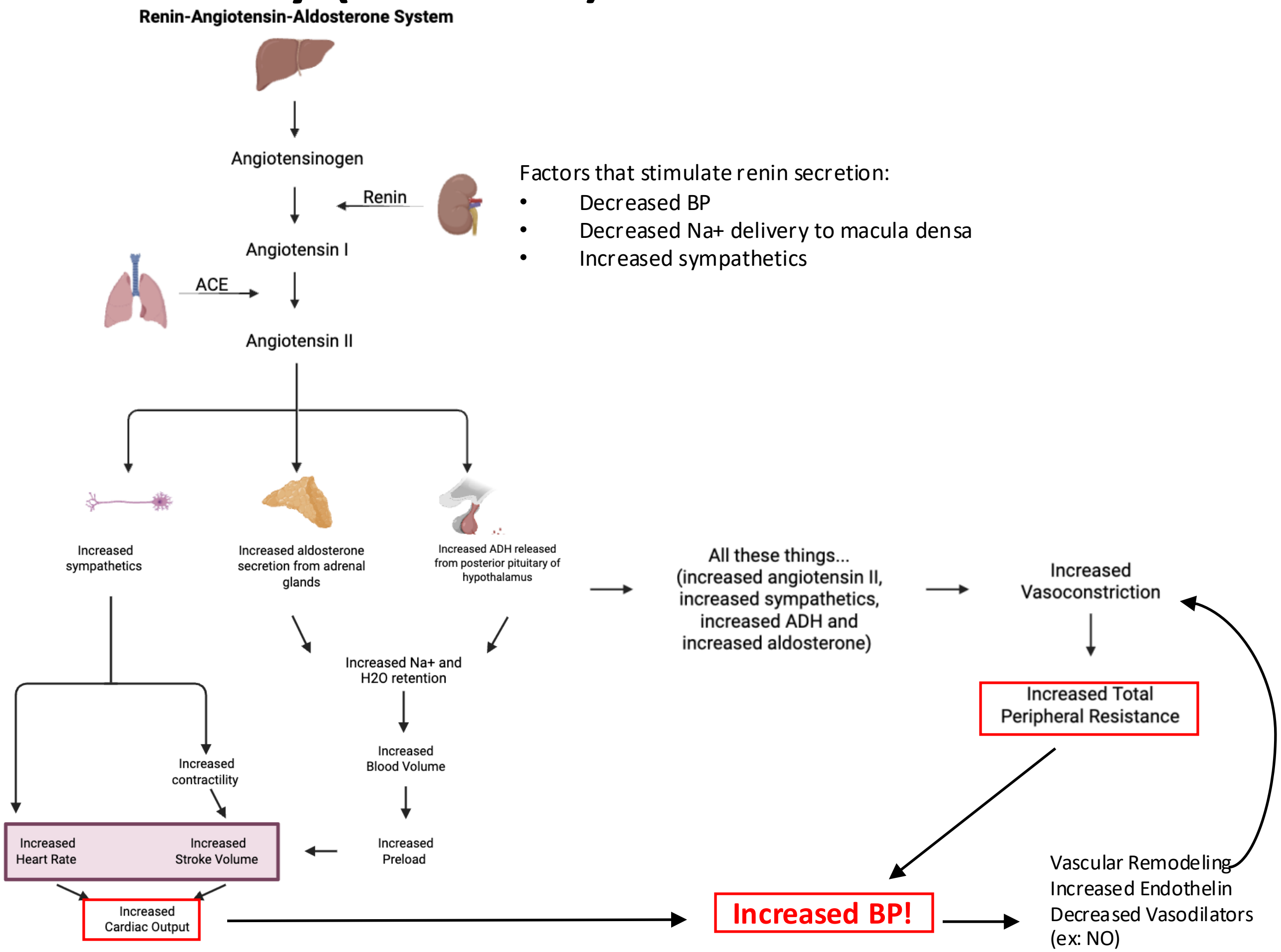
Types of Hypertension: Primary (Essential) HTN

- **Pathogenesis**: related to **increased Cardiac Output** or **increased Total Peripheral Resistance**

$$\boxed{MAP = CO \times TPR}$$

$$\downarrow$$

$$\boxed{CO = HR \times SV}$$



Types of Hypertension: Secondary HTN

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- **Remaining 10% of HTN; known underlying cause**
- **Consider in the following pts:**
  - HTN unresponsive to medications
  - Presents at young age
  - Older pts with new onset HTN
  - Acute worsening of previously well controlled BP
- **Causes:**

  Cushing's syndrome

  Hyperparathyroidism, hypothyroidism, hyperthyroidism

  Aortic coarctation (in young kids, but can be dx in adult)

  Meds: Oral Contraceptives, Chronic NSAID use, corticosteroids, decongestants; Illicit Drug Use (ex: cocaine)

  Pheochromocytoma

  Primary Aldosteronism

  Sleep Apnea

  Stenosis of renal artery and fibromuscular dysplasia

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Thinking back to our case #1…

What risk factors did our patient have and which are/are not modifiable?

- NSAID use (modifiable)
- Family hx of HTN (not modifiable)
- Sedentary lifestyle (modifiable)
- Fast food→ probably high in sodium (modifiable)
- Smoker (modifiable)
- Alcohol intake (modifiable)
- High BMI (modifiable)
- Race (not modifiable)

Q #1:

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A 49-year-old woman presents for an annual physical exam. She has no history of hypertension, diabetes, or cardiovascular disease. She exercises regularly and follows a healthy diet. She does not smoke or drink alcohol.

Vital signs are as follows: Blood pressure: 128/84 mmHg; Heart rate: 72/min; Respiratory rate: 12/min; Temperature: 36.9°C (98.4°F)

Repeat blood pressure readings on two subsequent visits are similar. Which of the following is the correct classification of this patient’s blood pressure?

A. Normal

B. Elevated

C. Hypertension Stage 1

D. Hypertension Stage 2

E. White coat hypertension

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Q #2:

A 60-year-old white male with PMHx of HTN, HLD, T2DM, and heavy tobacco use presents to clinic for a follow up for his high blood pressure. Despite escalating his anti-hypertensive regimen, he has persistently elevated blood pressure readings. His only reported symptom is a headache, and he denies chest pain. He is currently taking 4 antihypertensive medications. His BMI is 25 kg/m^2. Vitals are unremarkable except his blood pressure is 177/120 mmHg in the right arm and 179/123 mmHg in the left arm. Physical exam is consistent with an upper abdominal bruit. The finding is best explained by which of the following?

A. Aortic coarctation

B. Aortic dissection

C. Abdominal aortic aneurysm

D. Renal artery stenosis

E. Obstructive sleep apnea

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Q #3:

A 25-year-old female presents to the emergency department (ED) for an episode of epistaxis associated with extremely elevated blood pressure. Patient has no past medical history. Her blood pressure in the ED is noted to be 170/100 mmHg. Her BMI is 24 kg/m^2. On physical exam, there are no heart murmurs, abdominal bruits, or thyromegaly. Labs work up, including TSH, serum electrolytes, blood glucose, and creatinine, is normal. Which of the following additional tests would you order next?

A. Ultrasound of the renal arteries

B. Polysomnography for sleep apnea

C. Urine drug screen

D. Full thyroid panel

E. Lipid panel

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Evaluation of Hypertensive Patient

- **History: Go through those risk factors!**
  - Determine the presence of precipitating or aggravating factors
    - prescription medications, nonprescription NSAIDs, alcohol consumption
  - Duration of hypertension
  - Previous attempts at treatment
  - Signs of target-organ damage
  - Presence of other known risk factors for cardiovascular disease

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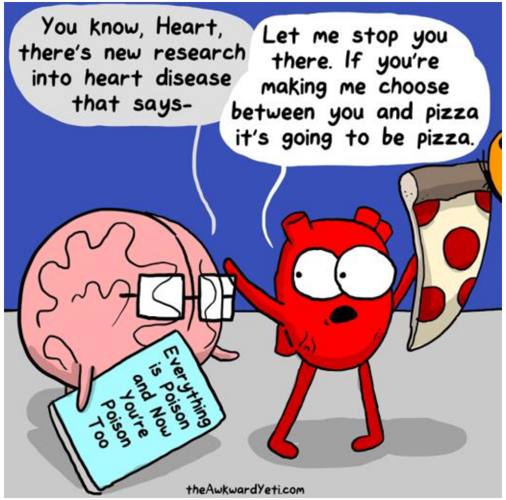

Evaluation of Hypertensive Patient

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Physical Exam:

- Many pts are often asymptomatic
- Evaluate for:

    signs of end-organ damage
    established cardiovascular disease
    evidence of potential causes of secondary HTN

General: body fat distribution, alertness

Fundoscopy: flame hemorrhages, arteriovenous nicking, papilledema, cotton wool spots (important funduscopic examination to evaluate for hypertensive retinopathy!)

Neck: carotid auscultation, neck circumference, thyroid

Heart: size, rhythm, sounds

i.e. S4 gallop – suggest LVH; systolic murmur (and HTN proximal and low BP distal) – coarctation

Lungs: rhonchi/wheezes, rales

Abdomen: renal mass (ADPKD), abdominal bruits (may suggest renal artery stenosis), femoral pulses

Extremities: peripheral pulses, edema

Neurologic: visual changes, weakness, confusion

Q #4:

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A 43-year-old male presents to the office for initial evaluation of high blood pressure. He exercises regularly and follows a low salt diet however his blood pressure continues to be around 154/86 mmHg in office, including today. He has no symptoms or past medical history. His BMI is 26 kg/m^2. On physical exam, no vascular bruits are heard, peripheral pulses are +2 and symmetrical, and heart and lung exams are unremarkable. ECG shows normal sinus rhythm. Complete blood count and complete metabolic panel are normal. Which additional study would you recommend for the patient?

A. 24-hour cortisol excretion

B. Lipid panel and urinalysis

C. Renal ultrasound

D. Aldosterone levels

E. Polysomnography

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Summary

- Understanding the definitions of hypertension and screening guidelines of a patient will allow for optimal treatment.
- Questions should be asked about the patients modifiable and non-modifiable risk factors as well as modification and drug use to guide the diagnosis of hypertension
- Diagnosing different types of hypertension is important for proper treatment.
- The physical exam should encompass multiple systems as complications of hypertension are systemic and cause devastating effects on health.